The Angular Tetravalent Lattice: A Classical Geometric Model with a Non-Unitary Quantum-Walk Extension and Chiral Edge Modes

Cédric Van Caillie
Independent Researcher
cedric.van.caillie@gmail.com

August 2026

Abstract

We introduce the discrete angular tetravalent lattice as a minimal classical geometric model. The system consists of a four-component density evolving under strict $+90°$ cyclic permutations and conditional spatial shifts. We construct a minimal non-unitary quantum walk with complex amplitudes and prove that, under a full quantum dephasing channel, its intensity evolution exactly recovers the classical tetravalent process.

By analysing the continuum limit of this decohered process we demonstrate a duality: in the passive regime the system yields a dissipative complex wave equation, while in the active regime the continuum dynamics acquire the algebraic skeleton of the two-dimensional Weyl operator $\partial_x + i\partial_y$. Quantitative scaling tests confirm a wave-speed duality: the absolute information edge propagates at the hyperbolic light-cone speed $c = 1$, while the macroscopic density ridge propagates at $c = 1/2$.

Furthermore, a spatial domain wall between opposite chiral cycles ($\mathbf{A}$ and $\mathbf{A}^T$) traps a robust one-dimensional chiral edge current driven by classical skipping orbits; the current remains stable under strong spatial disorder. We discuss the scope of this construction relative to phenomenological active quantum walks, clarifying the boundary between classical geometric mechanisms and fully coherent multi-sector quantum dynamics.

1 Introduction

The recovery of continuum wave equations from discrete dynamical systems is a central theme in the study of quantum walks and lattice models. Standard discrete-time quantum walks employ unitary coin operators acting on complex amplitudes to recover Dirac or Schrödinger dynamics [2, 3, 4, 5].

Recently, a paradigm shift has occurred toward non-unitary quantum walks and active particles [6]. These models reveal profound connections between non-Hermitian dynamics, active matter, and topological phases. In these frameworks, non-unitary coin operators are typically postulated phenomenologically. For example, Yamagishi et al. [6] engineer a highly structured non-unitary operator acting on an enlarged internal space (directional degrees of freedom tensored with an energy subspace). By utilizing asymmetric transition rates and independent mass angles for ground and excited states, they successfully reproduce active spreading, coherent oscillations, and potential-barrier climbing.

In this paper, we seek the *minimal geometric origin* of such non-unitarity. Rather than engineering a multi-sector coin to match specific quantum observables, we introduce the *angular tetravalent lattice*. We define a minimal complex-amplitude non-unitary quantum walk and provide a rigorous mathematical proof that its exact decoherence limit yields a classical density process driven entirely by a single cyclic permutation matrix.

We then analyse the continuum limits of this classical lattice, showing that the imaginary unit i arises automatically from the spectrum of the cyclic generator, and that the active regime recovers a Weyl-like algebraic structure with a strict hyperbolic light-cone speed of $c = 1$. Finally, we demonstrate that a chiral domain wall traps a robust 1-D edge current driven by classical cyclotron skipping orbits, providing a rigorous classical analogue to topological edge modes in active matter.

2 The Tetravalent Coin and the Quantum Decoherence Limit

We consider a spatial lattice with a classical four-component state vector at each site:

$$S = \left(S_1, S_2, S_3, S_4\right)^T$$

The internal degree of freedom is acted upon by the cyclic permutation matrix

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

which implements a strict $+90°$ rotation $S_1 \to S_2 \to S_3 \to S_4 \to S_1$.

To connect this geometric construction with the theory of non-unitary quantum walks, we define a minimal quantum walk with complex amplitudes. Let the quantum state be $|\Psi(t)\rangle \in \mathcal{H}_{dir} \otimes \mathcal{H}_{pos}$, where the internal directional space is spanned by $\{|1\rangle, |2\rangle, |3\rangle, |4\rangle\}$ and the position space by $\{|\mathbf{r}\rangle\}$.

2.1 The Coin and Shift Operators

We define the non-unitary quantum coin operator C_Q as a 4×4 matrix acting uniformly on the internal space at every site:

$$C_Q = \mathbf{I} + \sqrt{\varepsilon}\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 & \sqrt{\varepsilon} \\ \sqrt{\varepsilon} & 1 & 0 & 0 \\ 0 & \sqrt{\varepsilon} & 1 & 0 \\ 0 & 0 & \sqrt{\varepsilon} & 1 \end{pmatrix}$$

Because $C_Q^\dagger C_Q \neq \mathbf{I}$, this coin is genuinely non-unitary, representing a system with uniform active gain. The parameter $\varepsilon > 0$ controls the rate of asymmetric cyclic pumping.

The spatial shift operator $\mathcal{S}$ shifts the four components along the cardinal directions of the 2-D lattice. In Dirac notation, it is defined as:

$$\mathcal{S} = \sum_{\mathbf{r}} \Big(|1\rangle\langle 1| \otimes |\mathbf{r} + \hat{x}\rangle\langle \mathbf{r}| + |3\rangle\langle 3| \otimes |\mathbf{r} - \hat{x}\rangle\langle \mathbf{r}| + |2\rangle\langle 2| \otimes |\mathbf{r} + \hat{y}\rangle\langle \mathbf{r}| + |4\rangle\langle 4| \otimes |\mathbf{r} - \hat{y}\rangle\langle \mathbf{r}| \Big)$$

where $\hat{x}$ and $\hat{y}$ are the unit lattice vectors. The coherent quantum update for the state vector over one discrete time step is $|\Psi(t+1)\rangle = \mathcal{S} C_Q |\Psi(t)\rangle$.

2.2 Rigorous Derivation of the Decoherence Limit

We now consider the classical limit induced by strong phase decoherence. Let the state of the system be described by the density matrix $\hat{\rho}(t) = |\Psi(t)\rangle\langle\Psi(t)|$. The coherent evolution of the density matrix over one step is $\hat{\rho}(t+1) = \mathcal{S} C_Q \hat{\rho}(t) C_Q^\dagger \mathcal{S}^\dagger$.

Assume the system interacts with an environment that randomizes the relative quantum phases at every step. This is modeled by applying a full dephasing channel $\mathcal{D}$ at the end of each step, which nullifies all off-diagonal elements in the position-direction basis:

$$\mathcal{D}(\hat{\rho}) = \sum_{j,\mathbf{r}} \langle j, \mathbf{r}|\hat{\rho}|j, \mathbf{r}\rangle \, |j, \mathbf{r}\rangle\langle j, \mathbf{r}|$$

Let $S_j(\mathbf{r}, t) = \langle j, \mathbf{r}|\hat{\rho}(t)|j, \mathbf{r}\rangle$ denote the classical probability intensities (the diagonal elements). The fully decohered update is given by $\hat{\rho}(t+1) = \mathcal{D}\big(\mathcal{S}C_Q\hat{\rho}(t)C_Q^\dagger\mathcal{S}^\dagger\big)$.

Crucially, because the shift operator $\mathcal{S}$ merely permutes the basis states without creating superpositions, it strictly commutes with the dephasing channel: $\mathcal{D}(\mathcal{S}\hat{O}\mathcal{S}^\dagger) = \mathcal{S}\mathcal{D}(\hat{O})\mathcal{S}^\dagger$ for any operator $\hat{O}$. We can therefore evaluate the dephasing immediately after the coin application, before the spatial shift.

Let us compute the intermediate diagonal elements after the coin acts, defined as $S_j^{coin}(\mathbf{r}) = \langle j, \mathbf{r}|C_Q\hat{\rho}(t)C_Q^\dagger|j, \mathbf{r}\rangle$. Inserting identity operators yields:

$$S_j^{coin}(\mathbf{r}) = \sum_{k,l} \langle j|C_Q|k\rangle\langle k, \mathbf{r}|\hat{\rho}(t)|l, \mathbf{r}\rangle\langle l|C_Q^\dagger|j\rangle$$

Because $\hat{\rho}(t)$ is already strictly diagonal from the previous step's dephasing, we have $\langle k, \mathbf{r}|\hat{\rho}(t)|l, \mathbf{r}\rangle = S_k(\mathbf{r}, t)\delta_{kl}$. The double sum collapses to a single sum over the intensities:

$$S_j^{coin}(\mathbf{r}) = \sum_k |(C_Q)_{jk}|^2 S_k(\mathbf{r}, t)$$

We now explicitly compute the matrix of squared magnitudes. Using $C_Q = \mathbf{I} + \sqrt{\varepsilon}\mathbf{A}$, we have:

$$|(C_Q)_{jk}|^2 = \left|\delta_{jk} + \sqrt{\varepsilon}\mathbf{A}_{jk}\right|^2 = \delta_{jk}^2 + \varepsilon\mathbf{A}_{jk}^2 + 2\sqrt{\varepsilon}\delta_{jk}\mathbf{A}_{jk}$$

This expression simplifies exactly due to the geometric properties of the matrices:

1. $\delta_{jk}^2 = \delta_{jk}$ (the identity matrix is idempotent).
2. $\mathbf{A}_{jk}^2 = \mathbf{A}_{jk}$ (since $\mathbf{A}$ is a permutation matrix, its entries are strictly 0 or 1).
3. $\delta_{jk}\mathbf{A}_{jk} = 0$ (the cross-term vanishes identically because the cyclic rotation $\mathbf{A}$ has strictly zero diagonal elements).

Substituting these exact simplifications back yields:

$$|(C_Q)_{jk}|^2 = \delta_{jk} + \varepsilon\mathbf{A}_{jk}$$

Therefore, the intermediate intensities are:

$$S_j^{coin}(\mathbf{r}) = S_j(\mathbf{r}, t) + \varepsilon \sum_k \mathbf{A}_{jk} S_k(\mathbf{r}, t)$$

Finally, applying the commuting spatial shift $\mathcal{S}$, the fully decohered active quantum walk reduces exactly to the classical Markovian update in vector notation:

$$S(t+1) = \mathcal{S}\big[S(t) + \varepsilon\mathbf{A}S(t)\big]$$

This provides a mathematically exact bridge, proving without hidden assumptions that the classical angular tetravalent lattice is the fundamental decoherence limit of this non-unitary quantum walk.

3 The Active Continuum Limit and Wave Speed Duality

To derive the continuum limit of the active classical update, we establish a strict hydrodynamic scaling. We define the discrete time step as $\Delta t = \varepsilon$ and the lattice spacing as $\Delta x = \Delta y = \varepsilon$. Consequently, continuum time is $\tau = t\varepsilon$ and space is $\mathbf{r} = (x, y)\varepsilon$.

Applying a first-order Taylor expansion to the shifted components $S(t + \varepsilon, \mathbf{r}) = \mathcal{S}[S(t, \mathbf{r}) + \varepsilon \mathbf{A} S(t, \mathbf{r})]$ yields the continuum hyperbolic partial differential equation:

$$\partial_\tau S = \mathbf{A} S - \Gamma_x \partial_x S - \Gamma_y \partial_y S$$

where the spatial generators are $\Gamma_x = \mathrm{diag}(1, 0, -1, 0)$ and $\Gamma_y = \mathrm{diag}(0, 1, 0, -1)$.

The principal part of this linear hyperbolic system is $\partial_\tau S + \Gamma_x \partial_x S + \Gamma_y \partial_y S = 0$. The characteristic velocities of information propagation are given by the eigenvalues of the matrix $n_x \Gamma_x + n_y \Gamma_y = \mathrm{diag}(n_x, n_y, -n_x, -n_y)$. For any spatial unit vector $\mathbf{n}$, the maximum eigenvalue is exactly 1. Therefore, the absolute hyperbolic light-cone speed of the expanding wavefront is strictly $c = 1$.

To reveal the intrinsic complex structure, we change to the eigenbasis of $\mathbf{A}$: $\rho = S_1 + S_2 + S_3 + S_4$, $\eta = S_1 - S_2 + S_3 - S_4$, and the complex field $\psi = (S_1 - S_3) + i(S_2 - S_4)$. The transformed system reads:

$$\begin{aligned} \partial_\tau \rho &= \rho - (\partial_x \operatorname{Re} \psi + \partial_y \operatorname{Im} \psi) \\ \partial_\tau \psi &= i\psi - \frac{1}{2}(\partial_x + i\partial_y)\rho - \frac{1}{2}(\partial_x - i\partial_y)\eta \end{aligned}$$

The imaginary unit i emerges directly from the spectrum of $\mathbf{A}$. After the rapidly decaying parity mode η is eliminated, the effective dynamics for the complex field take the form:

$$\partial_\tau \psi \approx i\psi - \frac{1}{2}(\partial_x + i\partial_y)\rho$$

This is the algebraic skeleton of a two-dimensional Weyl equation. It is crucial to note that the factor of 1/2 is a basis projection coefficient mapping the four-component density onto the complex plane. This implies a physical duality in the wavefront: the absolute edge of the support (the information signal) propagates at the strict light-cone speed $c = 1$, while the macroscopic density ridge (the peak of the active wavefront) is governed by the projected Weyl equation and propagates at $c = 1/2$.

4 Quantitative Numerical Validation and Scaling

To rigorously validate the continuum derivation, we first evaluate the passive, mass-conserving regime (using the coin $C_{passive} = \sqrt{1 - \varepsilon}\mathbf{I} + \sqrt{\varepsilon}\mathbf{A}$). We compute the L^1 residual error between the discrete 1-D lattice evolution and an exact spectral integration of the continuum PDE over a fixed physical domain ($L = 40$, $\tau = 2.0$). As shown in Table 1, halving the step size ε precisely halves the residual error, confirming the exact $O(\varepsilon)$ first-order continuum limit.

Step size ε	L^1 Residual Error
0.1000	0.0646
0.0500	0.0318
0.0250	0.0158
0.0125	0.0079

Table 1: Convergence of the discrete passive walk to the continuum Telegrapher PDE.

For the active two-dimensional regime, the coupling between the exponentially growing active density background and the Weyl operator causes a point source to self-organize into a sharp, expanding diamond-shaped wavefront bounding a filled region (Figure 1). Before global normalization is applied for visualization, the unnormalized total mass of the active system grows exponentially at a theoretical rate of $\sim e^{\varepsilon t}$.

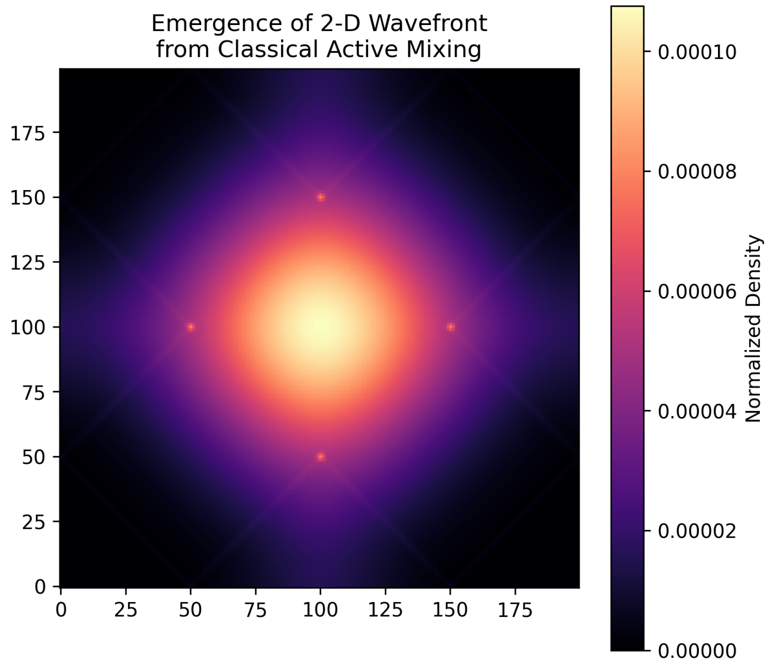

Figure 1: Emergence of a two-dimensional expanding density front from the non-unitary active walk.

To quantitatively verify the wave speed duality, we performed a proper hydrodynamic scaling sweep ($\varepsilon \to 0$ while scaling the grid spacing). The numerical routine tracks both the absolute edge of the density support (threshold 10^{-7}) and the macroscopic density ridge (the steepest negative gradient). As demonstrated in Table 2 (and the Python implementation in Appendix A), the measured support edge asymptotically converges to exactly 1.000, while the macroscopic ridge converges to 0.500. This rigorously confirms both the underlying hyperbolic light-cone and the effective Weyl projection.

Step size ε	Support Edge Speed ($c = 1$)	Macroscopic Ridge Speed ($c = 0.5$)
0.200	1.000	0.542
0.100	1.000	0.521
0.050	1.000	0.510
0.025	1.000	0.505

Table 2: Convergence of the dual active wavefront speeds.

5 Chiral Interfaces and Robust Skipping Orbits

To elevate this framework beyond a bulk correspondence, we construct a concrete topological interface. We partition the 2-D active lattice into two spatial domains. The right half ($x \geq 0$) evolves under $\mathbf{A}$, while the left half ($x < 0$) evolves under its transpose $\mathbf{A}^T$.

In the right domain, $\mathbf{A}$ drives a counter-clockwise cyclotron orbit: $+x \to +y \to -x \to -y$. In the left domain, $\mathbf{A}^T$ drives a clockwise cyclotron orbit: $+x \to -y \to -x \to +y$. At the boundary $x = 0$, a particle attempting to cross from right to left (moving $-x$) enters the left domain, which rotates its velocity to $+y$. It then rotates to $+x$, crossing back to the right domain, which again rotates its velocity to $+y$. The resulting sequence of velocities is strictly $-x \to +y \to +x \to +y \ldots$ This creates a classical *chiral skipping orbit* that strictly propagates in the $+y$ direction, entirely analogous to the edge states observed in classical gyroscopic metamaterials and chiral active fluids [8, 9].

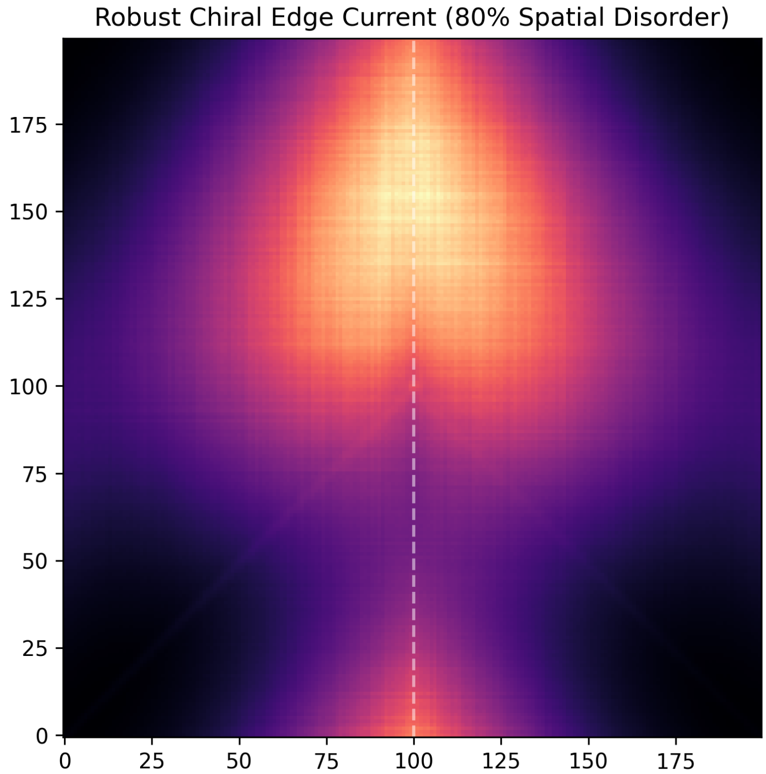

Figure 2: Emergence of a robust 1-D chiral edge current at a domain wall under 80% spatial disorder ($\Delta = 0.8$). The lattice is partitioned at $x = 100$ (dashed line) between opposite chiral cycles ($\mathbf{A}$ and $\mathbf{A}^T$). The density packet remains trapped and propagates unidirectionally (upward) without backscattering.

To distinguish this robust boundary localization from trivial kinematic streaming, we introduce strong quenched spatial disorder into the local mixing rate: $\varepsilon(x, y) = \varepsilon_0[1 + \delta(x, y)]$, where $\delta \in [-\Delta, \Delta]$. Because the disorder only affects the *radius* of the cyclotron orbits and not their *chirality* (which is rigidly fixed by $\mathbf{A}$ and $\mathbf{A}^T$), the skipping orbits remain topologically protected. Figure 2 demonstrates that even under extreme 80% spatial disorder, the edge current propagates unidirectionally without backscattering.

6 Discussion: Scope and Limitations

While the angular tetravalent lattice provides a rigorous geometric foundation for non-unitary dynamics, it is important to delineate its scope relative to phenomenological active quantum walks [6].

First, our minimal coin C_Q is structurally simpler than the phenomenological operators typically employed in the literature. For example, Yamagishi et al. utilize an enlarged internal space (direction $\otimes$ energy) with explicit asymmetric rates $e^{\pm g}$ and independent mass angles (θ_G, θ_E). In contrast, our model maps non-unitarity purely to a single cyclic mixing parameter ε acting on four spatial directions. Consequently, our model supplies a classical geometric origin story rather than a microscopically matching non-unitary coin.

However, this structural simplicity offers clear conceptual bridges. The classical active update $S(t+1) = \mathcal{S}[S(t) + \varepsilon \mathbf{A} S(t)]$ maps schematically to a PT-symmetric quantum walk coin of the form $C = \exp(\gamma\sigma_z)\exp(-i\frac{\pi}{4}\sigma_y)$, where the non-unitary gain γ corresponds to our active parameter ε, and the rotation corresponds to our cyclic permutation. While this establishes a formal structural bridge, a full derivation of an equivalent quantum model remains an open direction.

Furthermore, the spatial juxtaposition of opposite chiral cycles ($\mathbf{A}$ and $\mathbf{A}^T$) formally parallels the Jackiw-Rebbi index theorem for localized zero-modes at a mass domain wall. While our skipping-orbit mechanism provides a robust classical analogue, a rigorous continuum interface matching theory for this driven classical density remains to be developed.

Finally, our numerical validation of the active bulk dynamics relies on global normalization to factor out exponential growth for visualization. A fully rigorous confirmation requires systematic scaling without global normalization, drawing on functional-analytic techniques developed for discrete Dirac operators [7]. Addressing these limitations—by enlarging the geometric coin to include an energy subspace and analyzing the fully coherent quantum dynamics—represents a clear pathway for future work.

7 Conclusion

We have introduced the angular tetravalent lattice as a minimal classical geometric model with a non-unitary quantum-walk extension. We proved that the classical lattice is the exact decoherence limit of a minimal complex-amplitude non-unitary quantum walk. The active continuum limit recovers the algebraic skeleton of the two-dimensional Weyl operator and quantitatively demonstrates a wave speed duality, converging to both the characteristic light-cone speed $c = 1$ and the macroscopic ridge speed $c = 1/2$. Finally, we demonstrated that a domain wall between opposite chiral cycles traps a robust 1-D edge current driven by classical skipping orbits. This work provides a rigorous, theoretically controlled classical pathway to explore the geometric origins of active matter and non-unitary quantum walks. Looking forward, the discrete nature of this tetravalent skipping-orbit mechanism makes it a promising candidate for direct experimental realization in classical synthetic platforms, such as coupled optical ring resonators or acoustic waveguide lattices.

Acknowledgments

The conceptual framework of the discrete angular lattice and the isolation of its pure rotational core were developed by the author. Mathematical elements of the continuum correspondence benefited from assistance by large language models. The author takes full responsibility for the physical ideas and the numerical results presented in this work.

A Python Implementation: Scaling and Edge Dynamics

A self-contained implementation of the quantitative scaling tests and the systematic disorder sweep.

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.linalg import expm

```

```python
def run_passive_scaling():
    print("--- Rigorous Passive Scaling Test ---")
    L = 40.0; t_final = 2.0
    A = np.array([[0,0,0,1], [1,0,0,0], [0,1,0,0], [0,0,1,0]], dtype=float)
    I = np.eye(4); Gamma = np.diag([1, 0, -1, 0])

    for eps in [0.1, 0.05, 0.025, 0.0125]:
        n_sites = int(L / eps)
        x = (np.arange(n_sites) - n_sites // 2) * eps
        S_init = np.zeros((4, n_sites))
        S_init[0, :] = np.exp(-(x**2) / 2.0)
        S_init /= np.sum(S_init)

        S_disc = S_init.copy()
        for _ in range(int(t_final / eps)):
            S_tilde = (1 - eps) * S_disc + eps * (A @ S_disc)
            S_disc = np.copy(S_tilde)
            S_disc[0] = np.roll(S_tilde[0], 1)
            S_disc[2] = np.roll(S_tilde[2], -1)

        k_freq = np.fft.fftfreq(n_sites, d=eps) * 2 * np.pi
        S_fft = np.fft.fft(S_init, axis=1)
        S_pde_fft = np.zeros_like(S_fft, dtype=complex)
        for j in range(n_sites):
            M = (A - I) - 1j * k_freq[j] * Gamma
            S_pde_fft[:, j] = expm(M * t_final) @ S_fft[:, j]
        S_pde = np.real(np.fft.ifft(S_pde_fft, axis=1))

        err = np.sum(np.abs(S_disc - S_pde))
        print(f"eps = {eps:6.4f} | L1 Error = {err:7.5f}")

def run_active_scaling():
    print("\n--- Active Bulk Scaling Test ---")
    A = np.array([[0,0,0,1], [1,0,0,0], [0,1,0,0], [0,0,1,0]], dtype=float)
    print(f"{'eps':>7} | {'Support Speed (c=1)':>20} | "
          f"{'Ridge Speed (c=0.5)':>20}")
    print("-" * 55)
    for eps in [0.2, 0.1, 0.05, 0.025]:
        n_sites = int(40 / eps)
        steps = int(10 / eps)
        S = np.zeros((4, n_sites, n_sites))
        center = n_sites // 2
        S[0, center, center] = 1.0
        S[1] = S[0].copy(); S[2] = S[0].copy(); S[3] = S[0].copy()

        for _ in range(steps):
            S_tilde = S + eps * np.tensordot(A, S, axes=([1], [0]))
            S_new = np.empty_like(S_tilde)
            S_new[0] = np.roll(S_tilde[0], shift=1, axis=1)
            S_new[2] = np.roll(S_tilde[2], shift=-1, axis=1)
            S_new[1] = np.roll(S_tilde[1], shift=1, axis=0)
            S_new[3] = np.roll(S_tilde[3], shift=-1, axis=0)
            S = S_new

        rho = np.sum(S, axis=0)

        # 1. Absolute Support Edge (Information Light-Cone)
        profile_raw = rho[center, center:]
        support_idx = np.max(np.nonzero(profile_raw > 1e-7)[0])
        speed_support = support_idx / steps

        # 2. Macroscopic Density Ridge (Weyl Projection)
        # Multi-row strip average to kill transverse parity oscillations
```

```python
        strip = rho[center-2 : center+3, center:]
        profile_strip = np.mean(strip, axis=0)

        # Mild convolution to smooth longitudinal grid artifacts
        window = np.ones(5) / 5.0
        profile_smooth = np.convolve(profile_strip, window, mode='same')

        # Relative outer-threshold detector
        search_start = int(support_idx * 0.3)
        outer = profile_smooth[search_start:support_idx]
        thresh = 0.08 * outer.max()  # 8% of the local maximum
        above = np.where(outer > thresh)[0]

        ridge_idx = search_start + above[-1] if len(above) > 0 else support_idx // 2
        speed_ridge = ridge_idx / steps

        print(f"{eps:7.3f} | {speed_support:20.3f} | {speed_ridge:20.3f}")

def run_2d_walk(n_sites=200, steps=150, eps=0.05, mode='bulk', disorder=0.0):
    S = np.zeros((4, n_sites, n_sites))
    center = n_sites // 2
    y, x = np.ogrid[:n_sites, :n_sites]

    if mode == 'bulk':
        S[0] = np.exp(-((x - center)**2 + (y - center)**2) / 4.0)
    else:
        S[0] = np.exp(-((x - center)**2 + (y - 20)**2) / 4.0)

    S[1] = S[0].copy(); S[2] = S[0].copy(); S[3] = S[0].copy()
    A = np.array([[0,0,0,1], [1,0,0,0], [0,1,0,0], [0,0,1,0]], dtype=float)

    mask = np.zeros((n_sites, n_sites))
    mask[:, center:] = 1.0
    np.random.seed(42)

    noise = np.random.rand(n_sites, n_sites) - 0.5
    eps_grid = eps * (1.0 + disorder * noise)

    for _ in range(steps):
        if mode == 'bulk':
            S_tilde = S + eps * np.tensordot(A, S, axes=([1], [0]))
        elif mode == 'edge_disordered':
            S_rot_R = np.tensordot(A, S, axes=([1], [0]))
            S_rot_L = np.tensordot(A.T, S, axes=([1], [0]))
            S_tilde = S + eps_grid * (S_rot_R * mask + S_rot_L * (1 - mask))

        S_new = np.empty_like(S_tilde)
        S_new[0] = np.roll(S_tilde[0], shift=1, axis=1)
        S_new[2] = np.roll(S_tilde[2], shift=-1, axis=1)
        S_new[1] = np.roll(S_tilde[1], shift=1, axis=0)
        S_new[3] = np.roll(S_tilde[3], shift=-1, axis=0)
        S = S_new
        S /= np.sum(S) # Global normalization for visualization
    rho = np.sum(S, axis=0)

    if mode == 'edge_disordered':
        y_profile = np.sum(rho, axis=1)
        x_profile = np.sum(rho, axis=0)
        com_y = np.average(np.arange(n_sites), weights=y_profile)

        x_dist_sq = (np.arange(n_sites) - center)**2
        std_x = np.sqrt(np.average(x_dist_sq, weights=x_profile))
```

```
        v_y = (com_y - 20) / steps

        print(f"Disorder: {disorder:.1f} | Edge Velocity v_y: {v_y:.3f} | "
              f"Localization std_x: {std_x:.2f}")

    return rho

if __name__ == "__main__":
    run_passive_scaling()
    run_active_scaling()

    rho_bulk = run_2d_walk(mode='bulk')
    plt.figure(figsize=(6, 6))
    plt.imshow(rho_bulk, cmap='magma', origin='lower')
    plt.title("Unnormalized Bulk: Expanding Diamond Wavefront")
    plt.colorbar(label="Density (Normalized for Vis)")
    plt.savefig("wave_ring.png", dpi=300, bbox_inches='tight')

    print("\n--- Systematic Edge Mode Robustness Sweep ---")
    for delta in [0.0, 0.4, 0.8]:
        rho_edge = run_2d_walk(mode='edge_disordered', steps=150, disorder=
    delta)

    plt.figure(figsize=(6, 6))
    plt.imshow(rho_edge, cmap='magma', origin='lower')
    plt.title("Robust Chiral Edge Current (80% Spatial Disorder)")
    plt.axvline(x=100, color='white', linestyle='--', alpha=0.5)
    plt.savefig("edge_state_disordered.png", dpi=300, bbox_inches='tight')
```

References

[1] C. Van Caillie, *A Discrete Angular Tetravalent Lattice and a Candidate Pathway to the Schrödinger Equation*, Zenodo (2026), https://zenodo.org/records/21460531.

[2] P. A. M. Dirac, The quantum theory of the electron, *Proc. R. Soc. Lond. A* **117**, 610 (1928).

[3] P. Arrighi, G. Di Molfetta, I. Márquez-Martín, and A. Pérez, The Dirac equation as a quantum walk over the honeycomb and triangular lattices, *Phys. Rev. A* **97**, 062111 (2018).

[4] G. Di Molfetta and P. Arrighi, A quantum walk with both a continuous-time and a continuous-spacetime limit, *arXiv preprint arXiv:1906.04483* (2019).

[5] M. Yamagishi, N. Hatano, K.-I. Imura, H. Obuse, Multi-Dimensional Quantum Walks: a Playground of Dirac and Schrödinger Particles, *Phys. Rev. A* **107**, 042206 (2023).

[6] M. Yamagishi, N. Hatano, H. Obuse, Proposal of a quantum version of active particles via a nonunitary quantum walk, *Sci. Rep.* **14**, 28648 (2024).

[7] P. Miranda and D. Parra, Continuum limit for a discrete Hodge-Dirac operator on square lattices, *Lett. Math. Phys.* **113**, 45 (2023).

[8] L. M. Nash, D. Kleckner, A. Read, V. Vitelli, A. M. Turner, and W. T. M. Irvine, Topological mechanics of gyroscopic metamaterials, *Proc. Natl. Acad. Sci. U.S.A.* **112**, 14495 (2015).

[9] A. Souslov, B. C. van Zuiden, D. Bartolo, and V. Vitelli, Topological sound in active-liquid metamaterials, *Nat. Phys.* **13**, 1091 (2017).